Luis Renteria Lezano

Machine Learning Engineer | Data Pipelines | Applied AI (RAG & GenAI)

l_renterialezano@fanshaweonline.ca | 548-391-6575 | London, ON | luisrenteria.me | github.com/renteria-luis | linkedin.com/in/renteria-luis

PROFESSIONAL SUMMARY

Artificial Intelligence and Machine Learning graduate student (4.17/4.2 GPA, Dean's Honour Roll) who enjoys building machine learning systems from the ground up: designing them, training them, and shipping them as containerized FastAPI microservices with CI/CD. My work spans graph neural networks (GNN), retrieval-augmented generation (RAG), and agentic workflows, and I have a track record of shipping end-to-end applications using AI-assisted developer tools (vibe coding), from internal knowledge search tools to automated business logic. Background in Electronics Engineering: programming PLC control logic, designing HMIs, and running FAT/SAT validation on refinery automation projects, work that built real precision and a data-quality mindset he now carries into every pipeline.

EXPERIENCE

Technical Support Analyst — **Aug 2023 - Feb 2024**

Hikvision, Peru — *Hybrid*

- Tested pre-release hardware and software, identifying defects and providing structured user feedback on new features to product teams with exact reproduction steps and severity classification.
- Audited incoming equipment specifications against complex order records, executing strict line-by-line verification checks to prevent hardware mismatches.
- Delivered 20+ technical training sessions, in-person and online, to installers and integration partners across IP CCTV, alarms, access control, and video intercom systems (classes of 10-20), adjusting depth from high-level overview to granular configuration for the audience.

Applications Engineer — **Feb 2022 - Mar 2023**

Bermalar Consulting (Petroperu Refinery), Peru — *On-site*

- Programmed Allen-Bradley PLC control logic in Studio 5000 and designed the operator HMI from scratch in FactoryTalk View for a condensate treatment plant (activated carbon filters and mixed-bed units), owning the work end to end as one of two engineers.
- Executed strict FAT and SAT validation protocols against written specifications, logging every deviation for remediation and producing technical documentation to a controlled standard.
- Configured and calibrated Honeywell, Emerson, and Endress+Hauser field instruments on a HART/Enraf network, working directly with industrial sensor data at the point of acquisition for control-room monitoring.
- Produced and reviewed technical documentation (P&ID, single-line and multi-line diagrams, AutoCAD electrical and mechanical drawings), while maintaining legacy vendor toolchains on virtual machines to keep unsupported software running.

Renewals Specialist (Fortinet) — **Apr 2023 - Aug 2023**

TD Synnex, Peru — *Remote*

- Managed renewal cycles for Cisco and Fortinet service portfolios, coordinating with pre-sales engineering on technical solution scoping for enterprise clients.
- Collaborated with pre-sales teams and assisted enterprise clients with Fortinet and Cisco infrastructure, discusing technical requirements for network security and hardware deployments.

EDUCATION

Graduate Certificate in Artificial Intelligence and Machine Learning (Co-op) — **Sep 2025 - Present**

Fanshawe College, London, ON, Canada

- GPA: 4.17/4.2 | Dean's Honour Roll | Databases, Deep Learning, Natural Language Processing, ML Optimization, Data Visualization.

Bachelor's Degree in Electronic and Telecommunications Engineering — **Aug 2015 - Aug 2020**

Catholic University of San Pablo, Arequipa, Peru

- Awarded a full-scholarship Aerospace Engineering summer program at SJTU, China | Linear Algebra, Signals Processing, Control Systems

SKILLS

Programming & Core Tools: Python (Pandas, Pydantic, Regex), SQL, Git, Linux, Docker, Jira

Machine Learning & Deep Learning: PyTorch, scikit-learn, XGBoost, PyTorch Geometric, Graph Neural Networks (GNN), Hugging Face, Optuna, SHAP, Tensorflow

Applied AI & GenAI: Retrieval-Augmented Generation (RAG), chunking strategies, vector search (HNSW), agentic workflows, vibe coding, prompt engineering, local LLMs

MLOps & Backend: FastAPI (async microservices), GitHub Actions (CI/CD), Streamlit, model monitoring

Data Engineering & Analytics: Data validation, Data mapping (STTM, JSON, XML, CSV), Power BI, Advanced Excel (Power Query, XLOOKUP), exposure to foundational PySpark & Databricks

Industrial Control & Automation: Allen-Bradley PLC (Studio 5000), FactoryTalk View (HMI), field instrumentation (Honeywell, Emerson), HART, Modbus, AutoCAD, FAT/SAT validation

PROJECTS

Vehicle History Data Mapping Pipeline **Aug 2026 - Present**
Python, Pandas, Pydantic, SQL
github.com/renteria-luis/automotive-data-mapper

- Mapping heterogeneous JSON, XML, and CSV feeds into a canonical schema, establishing foundational data mapping rules.
- Investigating unmatched and ignored source data via strict Pydantic validation, routing malformed inputs to an exception queue for root cause analysis.
- Authoring mapping taxonomies and executed human-in-the-loop workflows, providing structured user feedback on new features for AI/ML labelling tools.

Cairn, On-Device RAG Study Assistant **May 2026 - Present**
EmbeddingGemma-300M, ObjectBox HNSW, local LLM, Android

- Designed the chunking and retrieval strategy for a fully offline Android app that answers questions over user documents (PDF, DOCX, PPTX, TXT), with no network calls, keeping private material on-device.
- Built sentence-aware chunking at 1600 characters with about 15% overlap, after fixed-window splitting was cutting sentences mid-thought, using character count as a proxy for the 512-token embedding limit, with a fixed-window fallback for oversized sentences.
- Specified notebook-scoped retrieval using EmbeddingGemma-300M embeddings (768-dim) over an ObjectBox HNSW index for accurate, hallucination-free on-device retrieval.

Fraud Detection Pipeline **Dec 2025 - Mar 2026**
Python, XGBoost, FastAPI, Docker, Hugging Face, GitHub Actions, Pydantic
github.com/renteria-luis/fraud-detection-v1

- Profiled and cleansed a 6.3M-record transaction dataset with a 349:1 class imbalance, removing transaction types with zero signal before modelling.
- Selected an operational classification threshold of 0.2226 to recover 6% of otherwise-missed fraud, documenting the precision-recall trade-off, and validated request payloads via Pydantic JSON schema validation.

Graph-Based AML Detection **Jun 2026 - Present**
PyTorch Geometric, GraphSAGE, FastAPI, Docker
github.com/renteria-luis/graph-aml-detector

- Building a money laundering anomaly detector on the Elliptic Bitcoin transaction graph (204K nodes, 166 features) using GraphSAGE using neighbor sampling.
- Using a temporal split instead of random sampling to prevent data leakage, and built an XGBoost tabular baseline to prove the graph structure adds value.

Wisp, Offline Quit-Smoking Companion App **Jun 2026 - Jul 2026**
Claude Code, AI Coding Assistant, Expo, React Native, TypeScript, Zustand, SQLite
github.com/renteria-luis/wisp

- Designed and shipped a fully offline, cross-platform quit-smoking app: adaptive onboarding routes each user onto a cold-turkey or gradual-reduction track, then builds a personalized, day-by-day reduction plan with no account, backend, or network calls.
- Directed the end-to-end build through agentic AI developer tools (vibe coding), personally designing the deterministic, fully tested rule engine and core logic, kept separate from the UI so the plan stays private and explainable.

Telco Churn Prediction **Nov 2025 - Jan 2026**
Python, Scikit-learn, XGBoost, ensembles, pandas, EDA
github.com/renteria-luis/telco-churn-prediction

- Built a soft-voting ensemble reaching 0.906 ROC-AUC and 87% recall at a deliberately tuned 0.3 threshold after removing leakage columns that restated the target, validating model choice at the real deployment threshold instead of the default 0.5 CV score.

House Prices Kaggle Competition **Oct 2025 - Nov 2025**
Python, Scikit-learn, ensembles, pandas, NumPy, Matplotlib
github.com/renteria-luis/house-prices-prediction

- Combined the tuned models into a Voting Regressor, cutting cross-validated RMSE to $20,341 from $23,130 for the next-best single model (Gradient Boosting), backed by a pytest suite validating feature-engineering output and pipeline shape.

LANGUAGES

English: Advanced | **Spanish:** Native | **French:** Basic